

IBM Quantum Composer: Hands-on Guide for Module 4

QML Lab Session

Objective

In this session, you will:

- Understand how qubits behave under different gates
- Build simple quantum circuits in IBM Quantum Composer
- Observe superposition, phase, interference, and entanglement
- Interpret probability plots and the Q-sphere
- Connect gate operations to the ideas behind Quantum Machine Learning

How This Demo Aligns with Module 4

This lab follows the same flow as the lecture module:

- **Quantum Information Theory** → deterministic evolution vs probabilistic measurement
- **Qubits: Operational View** → amplitudes, phase, and rotations
- **Quantum Gates** → X, H, Z, R_x , R_y , R_z
- **Multi-Qubit Systems and Entanglement** → CNOT and correlated outputs
- **Quantum Circuits** → order matters, interference, repeated sampling
- **Bridge to QML** → parameter tuning as the beginning of learning

Step 0: Open IBM Quantum Composer

Action:

- Log in to IBM Quantum Platform
- Open **Quantum Composer**
- Create a new circuit

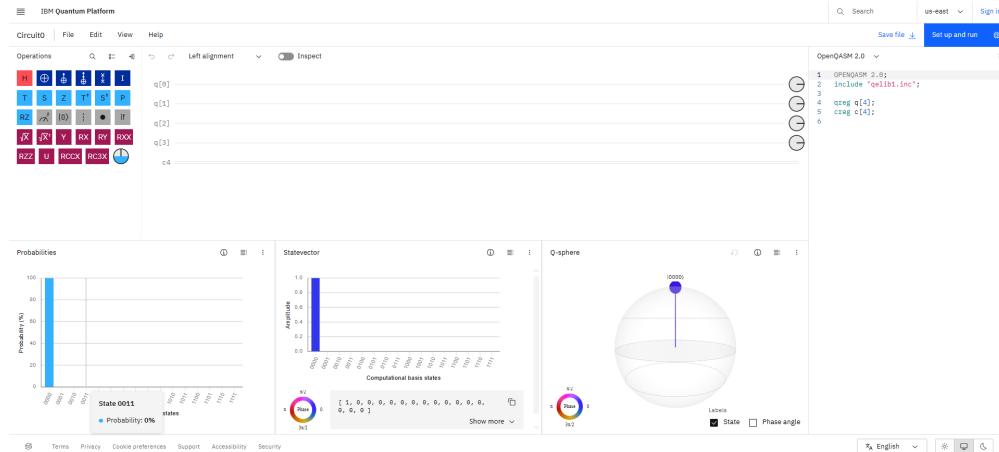
Observe the interface:

- Left panel: available gates

- Middle panel: circuit workspace
- Bottom / output area: probabilities or counts
- Right side / state view: Q-sphere or state visualization

Initial state:

$$|00\rangle$$



Step 1: Single-Qubit Superposition

Concept: Qubit as something we manipulate

Action:

- Start with a fresh circuit
- Apply $R_y(\pi/2)$ on $q[0]$
- Do not touch $q[1]$

Expected state:

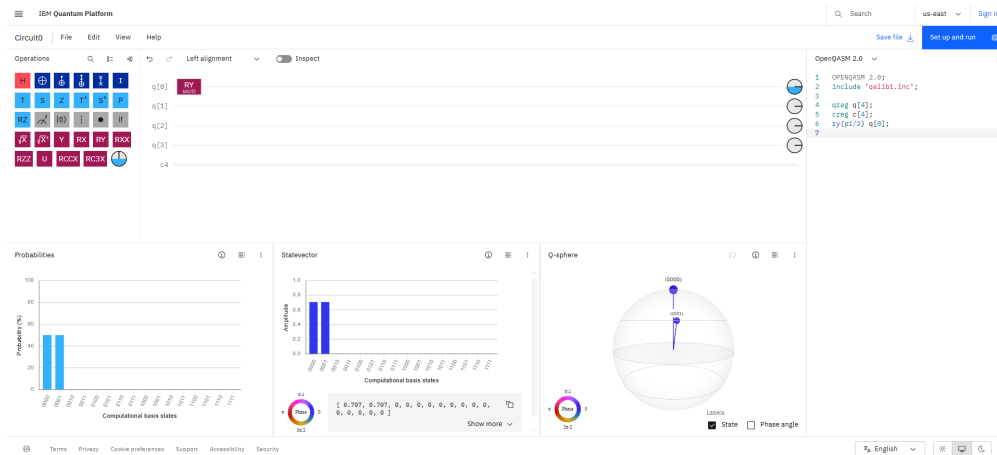
$$R_y(\pi/2)|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Observe:

- The first qubit moves into superposition
- Measurement of that qubit gives equal likelihood of 0 and 1
- Since the second qubit is unchanged, only two basis states appear overall

Learning:

- A single rotation can create superposition
- Gates change amplitudes, not just classical values



Step 2: Multi-Qubit Superposition

Concept: Multi-qubit state space grows quickly

Action:

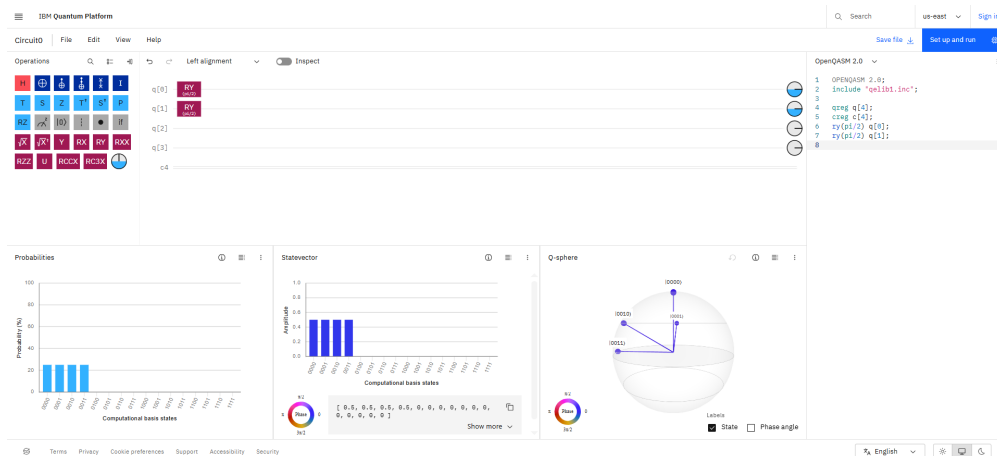
- Keep the previous circuit or start again
- Apply $R_y(\pi/2)$ on both $q[0]$ and $q[1]$

Observe:

- All four basis states appear
- If both rotations are $\pi/2$, the probabilities become equal
- Each of $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ appears with 25% probability

Learning:

- Multi-qubit systems represent many configurations together
- Two qubits already create a 4-dimensional state space



Step 3: Rotation Angle as a Control Knob

Concept: Rotation angle controls the output distribution

Action:

- Reset to a single-qubit circuit if needed
- Apply $R_y(\pi/4)$ on $q[0]$
- Run and observe
- Now change the angle to $R_y(\pi/8)$
- Run again
- Compare with $R_y(\pi/2)$

Observe:

- Probabilities are no longer equal for smaller angles
- The qubit is not “fully halfway” between $|0\rangle$ and $|1\rangle$
- Output distribution changes continuously with the angle

Learning:

- θ acts like a tunable parameter
- This is the earliest intuition for learnable quantum models



Step 4: X Gate (Bit Flip)

Concept: X gate behaves like a quantum NOT gate

Action:

- Reset the circuit

- Apply X gate on $q[0]$

Expected transformation:

$$|00\rangle \rightarrow |10\rangle$$

Observe:

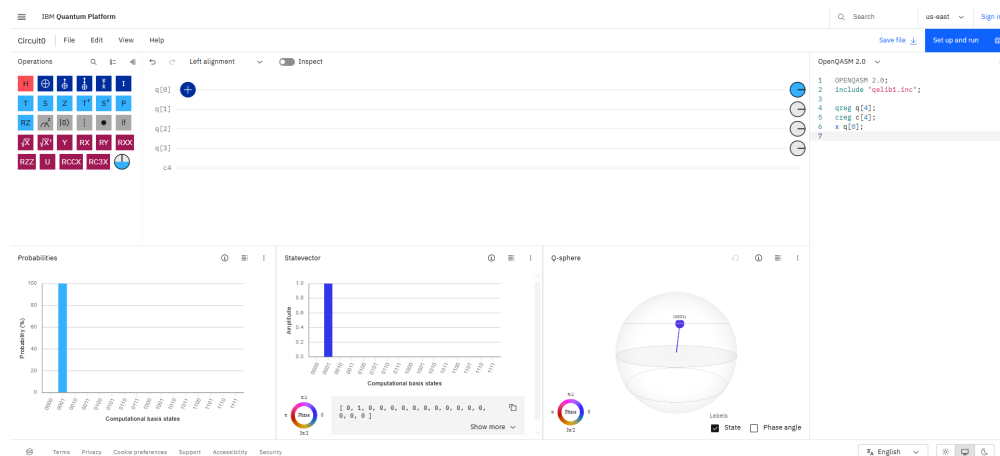
- The first qubit flips from 0 to 1
- Output becomes deterministic

Learning:

- X is the simplest gate to understand
- It shows how a quantum gate can still resemble a classical operation

Mini experiment:

- Apply X on $q[1]$ also
- Check whether the final output becomes $|11\rangle$
- Apply X twice on the same qubit and verify that you return to the original state



Step 5: H Gate and Z Gate — Phase is Hidden

Concept: Phase does not immediately show up in measurement probabilities

Action:

- Reset the circuit
- Apply H on $q[0]$
- Observe output probabilities
- Now apply Z on $q[0]$ after H
- Observe probabilities again

Expected logic:

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

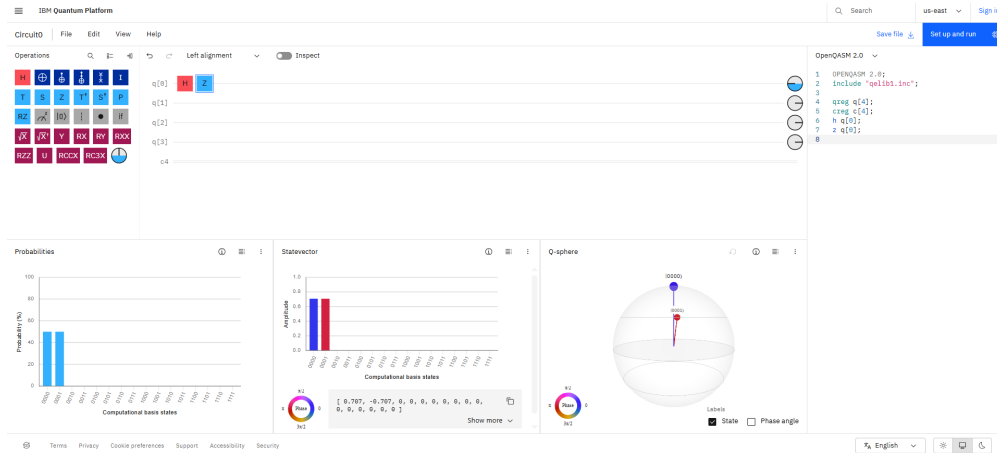
$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{Z} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Observe:

- The probability chart may look unchanged
- The Q-sphere / state visualization changes
- Phase has changed, but direct measurement in the computational basis does not immediately reveal it

Learning:

- Phase is a hidden resource
- Two states can have the same probabilities but behave differently later



Step 6: Making Phase Visible through Interference

Concept: Phase becomes visible after another gate

Action:

- Continue from the previous circuit or start fresh
- Build:

$$H \rightarrow Z \rightarrow H$$

on $q[0]$

Expected logic:

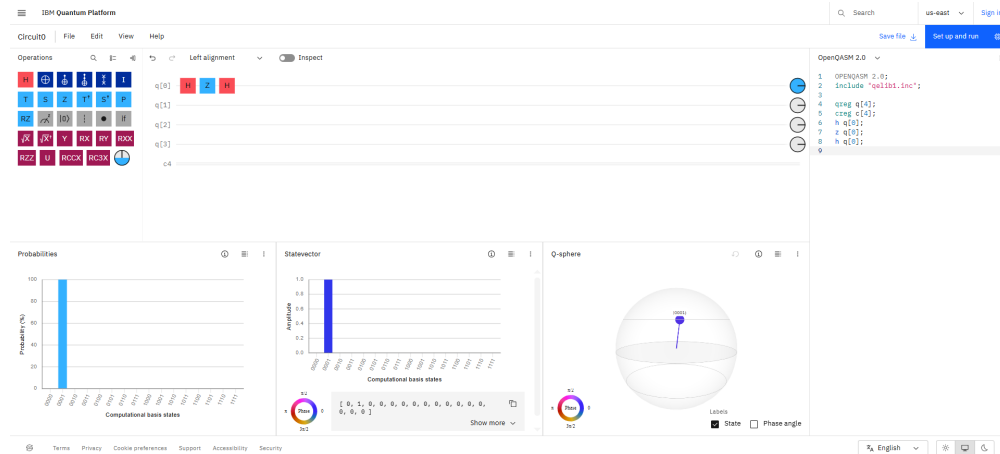
$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{Z} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \xrightarrow{H} |1\rangle$$

Observe:

- The output is no longer 50–50
- The hidden phase now changes the final measurement result

Learning:

- Phase affects interference
- Computation depends on how amplitudes combine after later gates



Step 7: R_x vs R_y vs R_z

Concept: Rotation axis matters

Action:

- Reset the circuit
- Apply $R_x(\pi/2)$ on $q[0]$
- Observe
- Repeat with $R_y(\pi/2)$
- Repeat with $R_z(\pi/2)$

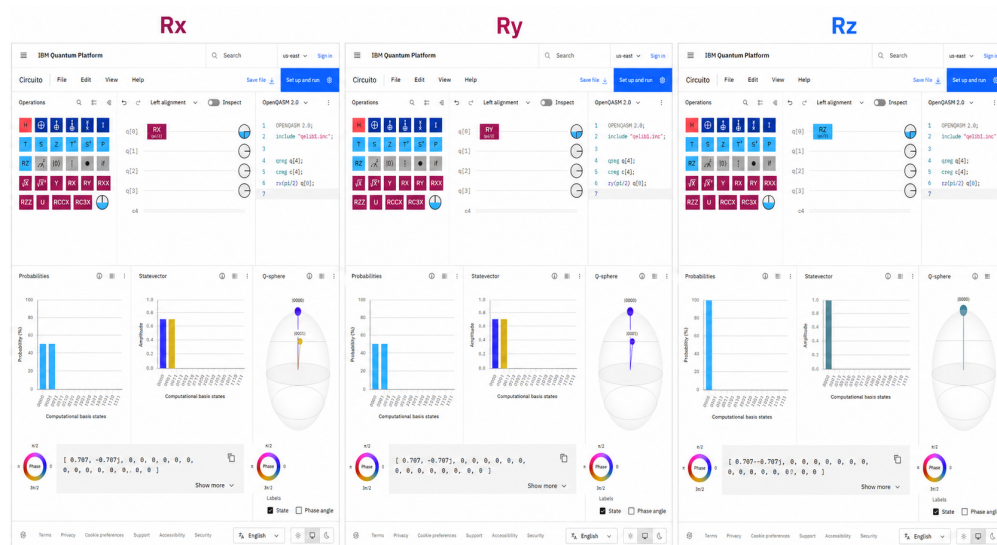
Observe:

- Different axes produce different states
- R_z often changes phase without directly changing measurement probabilities
- R_x and R_y can alter the visible probability distribution differently

Learning:

- Quantum computation is geometric
- Rotations are not all equivalent

- The axis of rotation affects the state trajectory on the Bloch sphere



Step 8: Entanglement using H + CNOT

Concept: Entanglement requires both superposition and interaction

Action:

- Reset the circuit
- Apply H on $q[0]$
- Apply CNOT with control = $q[0]$, target = $q[1]$

Expected logic:

$$|00\rangle \xrightarrow{H \text{ on } q[0]} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Observe:

- Only $|00\rangle$ and $|11\rangle$ appear
- The outputs are correlated

Learning:

- CNOT creates interaction between qubits
- H + CNOT produces an entangled Bell state

Step 10: Order Matters in Quantum Circuits

Concept: Same gates, different order, different result

Action:

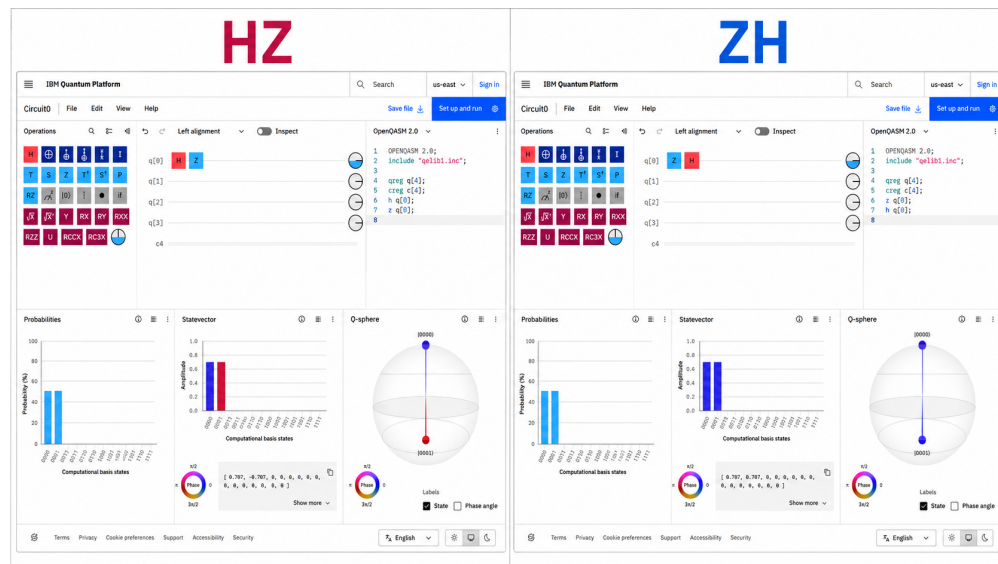
- Build Circuit A: $H \rightarrow Z$
- Build Circuit B: $Z \rightarrow H$
- Run both

Observe:

- Even though both circuits use the same gates, the results differ
- Quantum operations are order-sensitive

Learning:

- Quantum gates generally do not commute
- Circuit design is not just about gate choice, but also gate order



Step 11: Measurement Means Sampling

Concept: A circuit gives a distribution, not just one answer

Action:

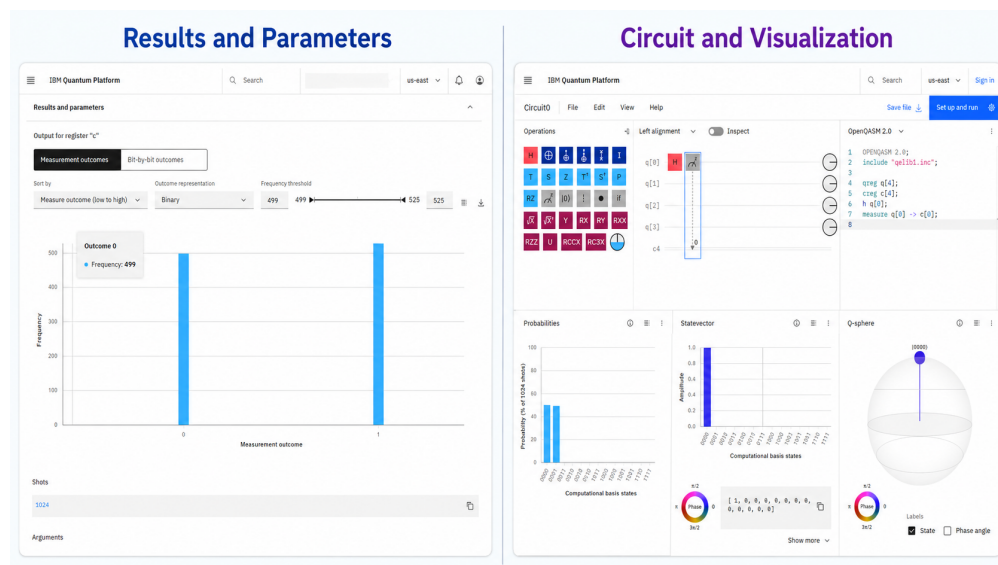
- Use any non-trivial circuit from earlier steps, such as H on one qubit
- Run it repeatedly with multiple shots
- Compare one-shot output with many-shot statistics

Observe:

- One run gives one outcome
- Many runs reveal the underlying probability distribution

Learning:

- Quantum circuits are understood statistically
- Measurement collapses the state, so repeated execution is necessary



Step 12: First Bridge to QML — Parameter Tuning

Concept: Learning in QML begins by tuning gate parameters

Action:

- Build a single-qubit circuit with $R_y(\theta)$ on $q[0]$
- Start with $\theta = \pi/8$
- Observe measurement probabilities
- Increase θ gradually
- Try to reach a circuit where the output is mostly $|1\rangle$

Goal:

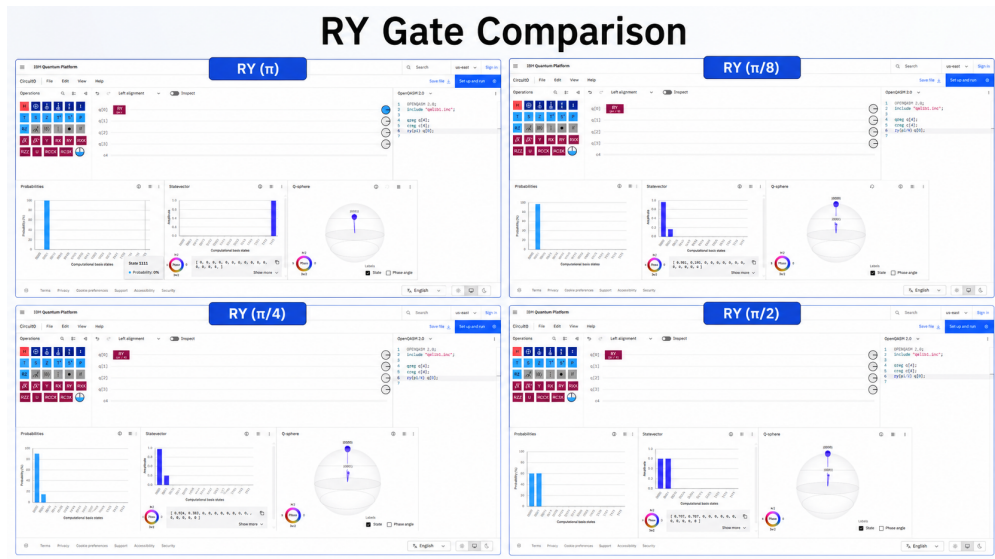
- Manually adjust θ to change the prediction behavior

Observe:

- Small changes in angle change the output distribution
- Tuning the angle changes how likely the system is to output 0 or 1

Learning:

- Rotation angles behave like model parameters
- This is the simplest intuition behind variational quantum circuits
- What you are doing manually here is what optimization algorithms do automatically later



Final Takeaways

- Gates change qubit states through controlled transformations
- Superposition enables multiple configurations to coexist
- Phase is hidden in direct measurement, but becomes visible through interference
- Entanglement creates correlations that cannot be explained classically
- Quantum circuits produce probability distributions, not just single answers
- Parameter tuning is the starting point of quantum machine learning